Heterogeneous Penalties at the Top: Glass Ceilings, Hours Convexity, and Motherhood Selection Among Highly Educated Women in the U.S.

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March 30, 2026

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Abstract

This paper extends Bach et al. (2018), who use Double Post-Lasso in a high-dimensional framework to document heterogeneity in the U.S. residual gender wage gap. The authors find that OLS substantially understates this heterogeneity: the median gap reaches 29% for less-educated women, while among college-educated women the penalty is selective, affecting only 40%. We then focus on White non-Hispanic women with a bachelor's degree — the group facing the largest documented penalties — and investigate three structural mechanisms. First, using a Causal Forest, we show that the penalty concentrates at the top of the earnings distribution, consistent with a glass ceiling. Second, using a Partially Linear Regression estimated via Double Machine Learning, we find that labour markets reward extreme hours convexly, but women working 70+ hours face an additional gender penalty of 20.9% that wipes out the market bonus entirely. Third, using a doubly robust DML-IRM estimator, we document a small motherhood wage premium of 2 to 5% among full-time workers, which we attribute to positive selection into employment rather than a causal return to motherhood. The structure of professional labour markets, not a lack of human capital, appears to be what holds highly educated women back. Why that structure penalises women the way it does remains unresolved.

1 Introduction

In 2023, full-time employed women in the U.S. earned 84 percent of the earnings of comparable men (U.S. Bureau of Labor Statistics, 2024). The gender wage gap is extensively documented and entails significant efficiency costs from the misallocation of human capital, beyond its serious implications for fairness. In the field of labour economics, research has focused on its determinants as a means to better understand the functioning of labour markets, the allocation of opportunities, and the effect of social structures on economic outcomes. Little attention has been given to understanding heterogeneity in the gender wage gap, which appears crucial to revealing the underlying mechanisms generating inequalities and to providing the necessary foundations for well-designed and targeted policy.

1.1 What Do We Know About the Gender Wage Gap?

The empirical literature has primarily studied the gender wage gap at the aggregate level using compositional methods. The Oaxaca-Blinder decomposition, the most widely used approach, decomposes average wage differences between men and women into a component stemming from differences in observable characteristics across genders and a component resulting from differential returns to observably similar men and women. While the first component can be estimated from observables — frequently including education, experience, occupation, firm type, or career interruptions — the second remains the 'unexplained' or 'residual' component. This term can capture discrimination; differences in unobservable factors such as productivity, ability, networks, motivation, bargaining power, and non-cognitive abilities; or relevant observable variables omitted due to limited understanding of labour market dynamics (Averkamp et al., 2024). Importantly, this approach ignores the fact that women are likely to be more selected than observably comparable men, given that they have faced fewer opportunities to arrive at the same situation as their male counterparts. The estimated residual gap is therefore, if anything, a lower bound on the true one.

Drawing on this method, Blau and Kahn (2017) document a decline in the gender wage gap in the U.S. from 1980 to 2010, largely driven by convergence in human capital between men and women. The unexplained gap fell sharply until 1989, stabilised over the succeeding two decades, and rose in relative importance. It is currently estimated to account for around 8 to 18% of the observed gap. Today, human capital factors explain little of the aggregate gap, while traditional factors such as industry and occupation remain the main drivers. The arrival of children translates into a motherhood penalty for women and a family premium for men. Several authors attribute the persistent residual gap to labour market discrimination.

2 The Paper We Extend

Only a few researchers have addressed heterogeneity in the residual gender wage gap. Goldin (2014) studies variation across multiple occupations and finds that inflexible occupations entail larger gender wage penalties. Other studies restrict their analysis to few, often arbitrarily selected, covariates and frequently fail to consider statistical significance (Bach et al., 2018). The paper we extend contributes to this strand of the literature by focusing on the variation of the gender wage gap across observational characteristics in a high-dimensional framework. The data-driven approach for model selection using Double Post-Lasso allows one to overcome simplistic approaches and to conduct valid statistical inference.

Drawing on data from the 2016 American Community Survey (ACS), Bach et al. (2018) estimate residual gender wage penalties across sixteen dimensions — capturing human capital factors, socioeconomic and household characteristics, family composition, demographic factors, geographic information, and labour market characteristics — for full-time and year-round employed individuals between 25 and 65 years old. Estimates are produced separately by education level.

2.1 Empirical Specification

The authors' estimation strategy draws on post-Lasso regression. Studying wage gap heterogeneity in a high-dimensional framework captures complex relations in the data while Lasso performs model selection by forcing some coefficients to zero, avoiding specification search. The authors assume approximate sparsity. A causal interpretation would require a conditional independence assumption, which the authors acknowledge is unlikely to hold; results are therefore purely descriptive.

Equation 1 presents the regression specification. Sixteen regressors interacted with gender allow the unexplained gap to vary across individual characteristics. Technical regressors consisting of all sixteen variables and their two-way interactions are included as controls, yielding a total of 2,068 and 4,382 regressors for the high school degree

and bachelor's degree subsamples, respectively. The parameter β_j captures the wage penalty associated with characteristic X_j between observationally comparable men and women.

$$\ln w_i = \alpha + \sum_{j=1}^{p_1} (\beta_j \cdot x_{i,j}) \cdot \text{gender}_i + z_i'\delta + \varepsilon_i \tag{1}$$

2.2 Double Post-Lasso: Valid Post-Model Selection Inference

Unless under considerably strong assumptions, inference on OLS coefficients selected by Lasso is likely to be invalid due to the possible omission of confounding factors. To overcome this, the authors draw on the double selection framework of Belloni et al. (2014a). Under approximate sparsity and restricted isometry, Double Lasso ensures first-order local insensitivity to nuisance parameters (Neyman orthogonality) and yields approximately unbiased and Gaussian estimators. The authors additionally build joint confidence bands using the multiplier bootstrap procedure of Chernozhukov et al. (2014, 2013a) combined with the valid joint test of Belloni et al. (2014b). Individual p-values are adjusted for multiple testing using the Romano–Wolf stepdown procedure (Romano and Wolf, 2005a,b, 2016), which controls the Family-Wise Error Rate.

3 Results from the Original Paper : OLS Conceals Heterogeneity: A Near-Universal Penalty for the Less-Educated, a Selective One for the College-Educated

Double Lasso estimation reveals pronounced heterogeneity in the gender wage gap that is largely obscured by OLS. While OLS suggests a homogeneous average residual gap of 17% (high school) and 14% (college-educated), the authors find that more than 90% of less-educated women face a significant penalty with a median of 29%, compared to only 40% of college-educated women with a median of 9%. Table 1 summarises the key determinants of this heterogeneity. Among these, the hours-worked result — whereby women working 70+ hours per week face a penalty of up to 11.86 pp — directly motivates the second extension. The counterintuitive motherhood premium among college-educated women with very young children motivates the third.

Table 1: Key results from Bach et al. (2018): Double Lasso estimates of the residual gender wage gap

Dimension	**High school**	**Bachelor's**	**Note**
Overall distribution			
OLS average gap	−17%	−14%	Homogeneous across women
Double Lasso median gap	−29%	−9%	OLS substantially understates
Share with significant penalty	>90%	≈40%	Gap is selective, not universal
Share with reversed gap	≈0%	≈4%	Absent in OLS estimates
Marital status (relative to single women)			
Married (spouse present)	−9.73 pp	−10.96 pp	
Parenthood			
Child aged ≤18	−5.07 pp	−5.31 pp	Motherhood penalty
Child aged ≤4	+2.89 pp	+8.09 pp	Selection into employment
Race & ethnicity (relative to White baseline)			
Black / African American	+4.37 to +8.19 pp		Smaller within-group gender penalty
Hispanic origin	n.s.		
Occupation (selected)			
Education / libraries	−18.36 pp	—	Largest penalty for HS women
Community / social services	+12.05 pp	—	Reduces gap for HS women
Healthcare practitioners	—	−4.07 pp	
Finance, insurance & real estate	large	large	Among largest in both groups
College major (bachelor's only, relative to education baseline)			
Engineering / architecture	—	+6.20 pp	Reduces gap
Computer / information science	—	+6.66 pp	Reduces gap
Math / statistics	—	+6.83 pp	Reduces gap
Law	—	−9.53 pp	Largest penalty among majors
Business	—	−6.21 pp	
Hours worked (relative to 35–40h baseline)			
More than 70 hours / week	−11.86 pp	−6.23 pp	Consistent with Goldin (2014)

4 Extensions: Structural Mechanisms Behind the Glass Ceiling

The results from the original paper motivate a deeper investigation into the structural mechanisms behind wage penalties among highly educated women. Our three extensions restrict attention to White non-Hispanic women with a bachelor's degree or higher — the group facing the largest documented residual penalties — and ask what specific combinations of career and life choices drive those penalties. This demographic restriction is also motivated by the selective nature of the gap among the college-educated: affecting roughly 40% of women rather than the near-universal 90% found among high school graduates, the penalty is concentrated among specific subgroups and career trajectories rather than uniformly distributed, making it better suited to structural analysis. The three extensions address this general question from complementary angles, each building on the findings of the previous one.

5 Extension 1: Heterogeneity and the Glass Ceiling

5.1 Motivation

The original paper establishes heterogeneity in the gender wage gap but does not specifically examine whether, among highly educated women, the penalty is concentrated at the top of the wage distribution — the defining feature of a glass ceiling. Prior literature suggests that gender wage gaps among high-skill workers are often largest for top earners (Bertrand et al., 2010), and that average estimates may hide large variation across women (Blau and Kahn, 2017). This extension therefore asks : **Who among highly educated White non-Hispanic women actually faces the wage penalty, and is it disproportionately borne by those at the top of the earnings distribution?**

5.2 Parametric Estimation: Double Post-Lasso

Replicating the authors' interacted regression model on the restricted subsample, Double Post-Lasso estimates are presented in Table 3. The results confirm that the largest wage gaps are driven by sector of employment. Penalties are most pronounced in traditionally masculine occupations and industries — finance, computer science, mathematics, and physical sciences — and in corresponding degree fields. Women working more than 70 hours per week earn approximately 6.5% less than comparable men. Marriage and parenthood each carry a penalty of approximately 4–5% relative to comparable single, childless women, consistent with the broader literature.

5.3 Non-Parametric Estimation: Causal Forest

We complement the parametric approach with a Causal Forest (Wager and Athey, 2018; Athey and Wager, 2019) to estimate the Conditional Average Treatment Effect (CATE) of being a woman on earnings non-parametrically. The Causal Forest first reduces confounding through nuisance models to obtain residualized outcome and treatment variables, clusters observations to maximise treatment effect heterogeneity, constructs local adaptive weights, and estimates the CATE by solving a local weighted orthogonal moment restriction. We implement it using the `grf` package in R, which uses regression trees for nuisance estimation, constructs the forest using honest cross-fitting, and obtains standard errors via bootstrap of little bags. All code for this and subsequent extensions is available in our GitHub repository.[1]

5.3.1 Heterogeneous Penalties: Half of Highly Educated Women Face No Significant Gap

Individual CATE estimates, shown in Figure 1a, confirm pronounced heterogeneity. Half of the individuals in the sample present an estimated gap that is large and statistically significantly below zero, ranging from 3% to 43.69%, while for the other half the estimate is not significantly different from zero. Feature importance analysis (Table 5) identifies the primary drivers of this heterogeneity as family structure — particularly having children and being married — followed by industry, with finance standing out as a sector where the gap behaves markedly differently from others. Age and experience also predict heterogeneity, suggesting the gap evolves over careers, though their importance is secondary to marital and parental status.

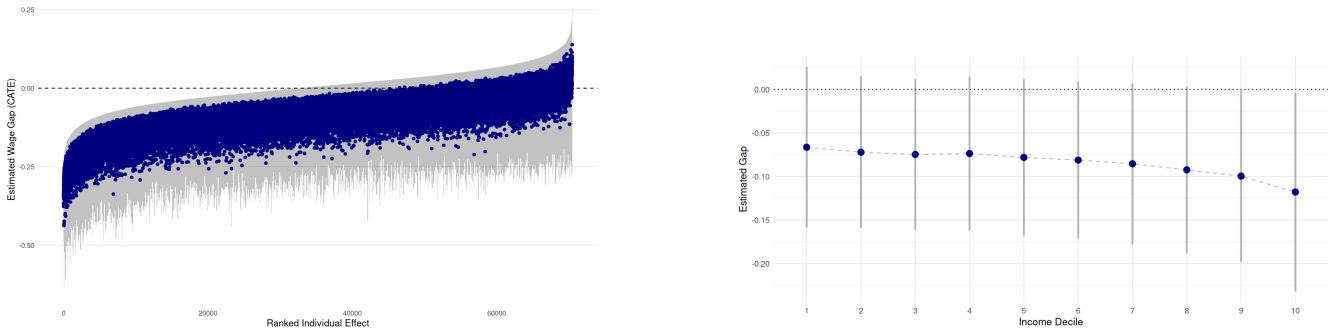
5.3.2 Glass Ceiling: Gap by Income Decile

Figure 1b shows that the average estimated wage gap widens as income increases. The gap is fairly stable across the lower and middle deciles at around 7.5%, then increases sharply above the median, reaching an average of 11.79% for the top income decile — with most estimates significantly different from zero. **This constitutes direct evidence of a glass ceiling: the wage penalty is not uniformly distributed but is disproportionately borne by women at the top of the earnings distribution**.

5.3.3 Extreme Hours and Masculine Sectors Drive the Largest Penalties

Best Linear Projection of the CATE onto the covariate space (Table 4) confirms the Double Post-Lasso results. The most important predictor of individual gap severity is working more than 70 hours per week, associated with an average penalty of 21% relative to comparable men. Women are penalised in traditionally masculine sectors while holding a slight advantage in a range of other sectors, though the magnitude of penalties consistently exceeds that of advantages. Married women and mothers face gaps of around 4–5%, consistent across both estimation methods.

[1] https://github.com/selaazdoudi/Heterogeneous-Penalties-at-the-Top/tree/main



(a) Individual Wage Gap Estimates and 95% CI

(b) Average Wage Gap by Income Decile

Figure 1: Causal Forest: Heterogeneity and Glass Ceiling

6 Extension 2: The Flexibility Price and Convex Returns to Hours

6.1 Motivation

Extension 1 establishes that the wage penalty among highly educated White non-Hispanic women is concentrated at the top of the income distribution, and that working more than 70 hours per week is the single strongest predictor of individual gap severity. This pattern calls for a structural explanation. A natural candidate is the mechanism identified by Goldin (2014): professional labour markets exhibit convex returns to hours, meaning the hourly wage increases more than proportionally as workers supply extreme hours. Formally:

$$\frac{\partial^2 \ln w}{\partial h^2} > 0.$$

Under this structure, workers who credibly commit to supplying 60 or 70+ hours per week receive a disproportionately high hourly rate — a *flexibility price* that penalises those who cannot sustain such hours regardless of total productivity. This raises a precise empirical question: **Do women receive the same returns to extreme hours as men, or do they face an additional gender-specific penalty at the high-hours threshold that cancels out the market bonus?** Our hypothesis is that both effects are present simultaneously — the earnings schedule is convex for everyone, but women who supply extreme hours face an extra penalty relative to comparable men that largely offsets the premium. If this is the case, The glass ceiling, in this reading, would be less about who gets promoted and more about who gets paid for working extreme hours.

6.2 Empirical Strategy: Partially Linear Regression via Double Machine Learning

To test this hypothesis, we estimate for each hours category $h \in \{50\text{–}59\text{h}, 60\text{–}69\text{h}, 70\text{h}+\}$ the wage premium associated with belonging to that category relative to the 40–49 hour baseline, controlling for a rich set of worker characteristics. We use a **Partially Linear Regression (PLR)** model estimated via Double Machine Learning (Chernozhukov et al., 2018):

$$Y_i = \theta_h D_{ih} + g_{0h}(X_i) + \varepsilon_i, \tag{2}$$

where Y_i denotes log weekly wages, D_{ih} is an indicator for hours category h, and X_i is a high-dimensional vector of controls including occupation, industry, college major, marital status, parenthood, region, metropolitan status, and labour market experience. The parameter θ_h captures the wage premium for category h relative to the baseline after partialling out all observable confounders.

Identification proceeds through residualization. Nuisance functions $\hat{g}_h(X_i) \approx E[Y_i \mid X_i]$ and $\hat{m}_h(X_i) \approx E[D_{ih} \mid X_i]$ are estimated using random forests (500 trees), yielding residualized variables $\tilde{Y}_i = Y_i - \hat{g}_h(X_i)$ and $\tilde{D}_{ih} = D_{ih} - \hat{m}_h(X_i)$. The estimator $\hat{\theta}_h$ solves $\frac{1}{n}\sum_{i=1}^{n} \tilde{D}_{ih}(\tilde{Y}_i - \hat{\theta}_h \tilde{D}_{ih}) = 0$ and satisfies Neyman orthogonality:

$$\frac{\partial}{\partial \eta} E[\psi(W_i; \theta_0, \eta)]\Big|_{\eta=\eta_0} = 0,$$

guaranteeing that small errors in nuisance estimation have only second-order effects on $\hat{\theta}_h$. Combined with 5-fold cross-fitting to prevent overfitting bias, this yields approximately unbiased and Gaussian estimators. The implementation uses the `DoubleML` package in R (Bach et al., 2024) with `mlr3` and `mlr3learners` providing the random forest learners. Replication code is available in our GitHub repository.[2]

6.3 Results

Figure 2 presents the estimated market premium for each hours category alongside the gender-specific penalty from the Causal Forest best linear projections of Extension 1.

6.3.1 Convex Market Returns to Hours

The market premium for long hours is large and convex: working 50–59 hours per week is associated with a premium of approximately +20.5% relative to the 40–49 hour baseline, rising to +26% at 60–69 hours and remaining at a similar level beyond 70 hours. This confirms the presence of significant non-linearity in the earnings schedule of professional labour markets, rewarding extreme labour supply disproportionately.

6.3.2 The Gender Penalty at the Extreme Tail

Women who supply 70 or more hours per week face an additional gender-specific penalty of approximately –20.9% relative to comparable men working the same hours. The net effect for women at extreme hours is therefore close to zero: the market bonus is almost entirely offset by the gender penalty that emerges at this threshold. Women working 50–69 hours experience only small or statistically negligible gender penalties, so they do benefit from the market premium to a meaningful degree. It is specifically at the extreme tail of the hours distribution that the glass ceiling manifests.

These results are consistent with the flexibility price hypothesis. The wage gap at the top of the distribution documented in Extension 1 is not simply a product of occupation or sector sorting: it is amplified by a structural feature of earnings schedules whereby the highest-paying work arrangements carry an additional gender penalty that prevents women from fully appropriating the associated returns.

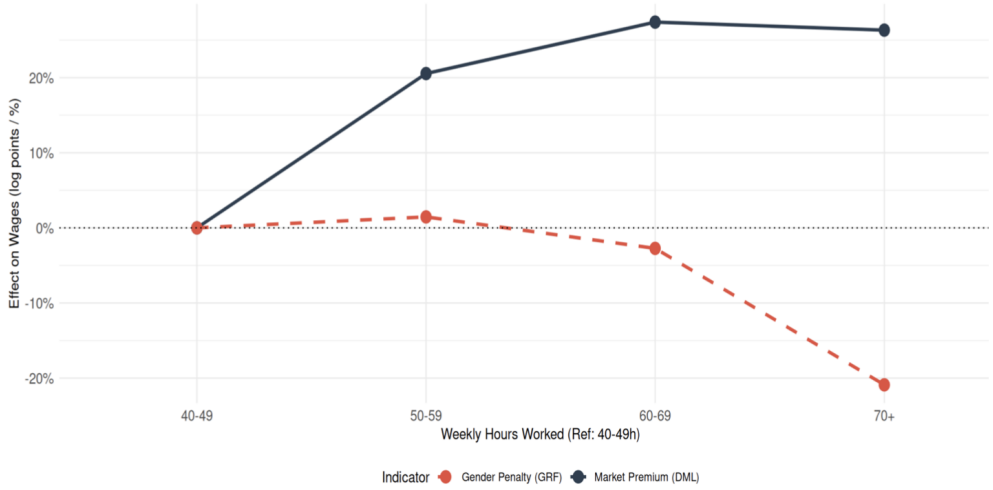


Figure 2: Market premium and gender penalty by hours worked category.

[2] https://github.com/selaazdoudi/Heterogeneous-Penalties-at-the-Top/tree/main

7 Extension 3: Motherhood Premium or Selection Artefact?

7.1 Motivation

The starting point for this extension is a striking finding from the original paper: among college-educated women, those with children aged four or younger appear to face a *smaller* gender wage gap than comparable childless women, and may even display a slight wage premium. This is surprising given the extensive literature documenting a motherhood penalty — the well-established finding that mothers earn less than comparable childless women, with estimates typically ranging from 4 to 13% per child in the United States (Waldfogel, 1997; Budig and England, 2001; Gangl and Ziefle, 2009). Our goal is to test whether this apparent premium survives once mothers and non-mothers are rendered comparable on a rich set of observable characteristics, or whether it is an artefact of selection into full-time employment.

Because the dataset is restricted to full-time, year-round workers, the analysis cannot directly address selection out of employment — women who left the labour market after childbirth are not observed. The parameter we estimate is therefore conditional on remaining employed full time:

$$\theta^{FT} = E[Y(1) - Y(0) \mid FT = 1],$$

where Y denotes log weekly wages and $D = 1$ indicates motherhood (having a child in the household). The empirical question is thus whether, among women who remain strongly attached to the labour market, mothers earn more or less than comparable non-mothers.

7.2 Empirical Strategy: DML Interactive Regression Model

We use a **Double Machine Learning Interactive Regression Model (DML-IRM)** estimator. Mothers and non-mothers differ simultaneously along many dimensions — age, labour market experience, marital status, occupation, industry, region, metropolitan status, and hours worked — making a simple mean comparison misleading. The empirical strategy relies on the conditional independence assumption $(Y(1), Y(0)) \perp D \mid X$. The estimation is implemented in R using the `DoubleML` package (Bach et al., 2024), with `mlr3` and `mlr3learners` providing the underlying learners. Penalized logistic regression via `glmnet` is used for both the propensity score and the outcome regressions. Balance diagnostics are produced using the `cobalt` package.[3]

7.2.1 Propensity Score Weighting

We first estimate the propensity score $e(X) = P(D = 1 \mid X)$ using penalized logistic regression, and construct inverse-probability weights:

$$w_i = \frac{D_i}{e(X_i)} + \frac{1 - D_i}{1 - e(X_i)},$$

which rebalance the sample so that mothers and non-mothers become comparable on observables. Balance diagnostics confirm that standardized mean differences fall below 0.11 across all covariates after reweighting.

7.2.2 Doubly Robust Estimation and Common Support

The final treatment effect is estimated via a doubly robust DML-IRM procedure combining the propensity score $P(D = 1 \mid X)$ and outcome regressions $E[Y \mid D, X]$ in an orthogonal score, with cross-fitting used to limit overfitting bias. To address limited overlap in the propensity score distribution, we estimate the model under three trimming rules — $0.02 < e(X) < 0.98$, $0.05 < e(X) < 0.95$, and $0.10 < e(X) < 0.90$ — restricting attention to regions of common support where mothers and non-mothers are genuinely comparable.

7.3 Motherhood Is Associated with a Small Wage Premium Among Strongly Attached Workers

Table 2 reports the DML-IRM estimates of the motherhood effect under alternative trimming rules. The estimated premium is positive and statistically significant across all specifications, and grows larger as the trimming rule becomes stricter — consistent with the effect being strongest where mothers and non-mothers are most comparable on observables.

[3] https://github.com/selaazdoudi/Heterogeneous-Penalties-at-the-Top/tree/main

Table 2: DML-IRM estimates of the motherhood wage effect under alternative trimming rules

Trimming rule	**Estimate**	**Effect (%)**	**Treated ESS**
No trimming	0.0191	+1.93%	—
2% trimming $(0.02 < e(X) < 0.98)$	0.0184	+1.85%	≈4,255
5% trimming $(0.05 < e(X) < 0.95)$	0.0328	+3.33%	≈7,129
10% trimming $(0.10 < e(X) < 0.90)$	0.0500	+5.13%	—

Outcome: log weekly wages. Treatment: having a child in the household ($D = 1$).
Nuisance models estimated via penalized logistic regression with 5-fold cross-fitting.
ESS: effective sample size of the treated group after reweighting.

Among highly educated women who remain in full-time employment, motherhood is thus associated with a small but robust wage premium of between 1.9% and 5.1% depending on the comparison group. The fact that the premium grows under stricter trimming suggests it is strongest among women who are most similar to each other on observables — that is, where the conditional independence assumption is most credible. This result should not be interpreted as a causal effect of motherhood: it likely reflects positive selection into full-time employment, whereby mothers who remain strongly attached to the labour market after childbirth are more productive, more career-committed, or hold positions offering higher wages to offset childcare costs..

8 Conclusion

This paper extends Bach et al. (2018) to show that the gender wage gap is far more heterogeneous than average estimates suggest, and that its structure among highly educated women points to specific structural mechanisms rather than uniform discrimination.

The authors find a median residual gap of 29% for less-educated women — nearly double the OLS estimate — while among college-educated women the gap is selective, concentrated among specific occupations, sectors, and family configurations. Our three extensions, restricted to White non-Hispanic women with a bachelor's degree, sharpen this picture considerably. The penalty is a glass ceiling in the literal sense: it is disproportionately borne by women at the top of the earnings distribution, with the top decile facing an average gap of 11.79%, driven primarily by finance, long hours, and family formation. Labour markets reward extreme hours convexly, but women working 70+ hours face an additional gender penalty of 20.9% that wipes out the market bonus entirely — they put in the hours but do not collect the returns. Among strongly attached mothers, motherhood is associated with a small wage premium of 2 to 5%, which grows stronger under stricter comparison of mothers and non-mothers.

Three limitations bear directly on the interpretation of these findings. First, none of the estimates in this paper — neither in the discussion of Bach et al. (2018) nor in the extensions — carry a causal interpretation. A causal reading would require the conditional independence assumption to hold: that, conditional on the observable covariates X, the treatment is as good as randomly assigned. This is unlikely to hold in a cross-sectional survey setting, where unobserved differences in preferences, productivity, or bargaining power plausibly confound all estimates. The results are therefore best read as descriptive — characterising the structure of residual wage differences rather than identifying their causes.

Second, our test of the flexibility price hypothesis in Extension 2 relies on self-reported hours worked, which are recorded in discrete brackets (50–59h, 60–69h, 70h+) rather than as a continuous variable. This limits our ability to trace the precise shape of the hours-earnings relationship and to pinpoint exactly where the gender penalty emerges within the distribution of extreme hours. Continuous hours data, or finer-grained brackets, would allow a more precise test of whether convexity and the gender penalty grow together smoothly or whether the penalty is truly concentrated at a specific threshold.

Third, the motherhood premium documented in Extension 3 should be interpreted with particular caution. Because the dataset is restricted to full-time, year-round workers by construction, mothers who reduced their hours or left employment after childbirth are entirely absent from the sample. The estimated premium is therefore conditional on a strong and non-random selection into continued full-time work: the mothers we observe are likely those with the strongest labour market attachment, the most career-compatible occupations, or access to childcare arrangements that make full-time employment feasible. The premium almost certainly reflects this selection rather than a genuine wage return to motherhood. Isolating the true effect would require panel data tracking women

before and after childbirth, or quasi-experimental variation in childcare access, neither of which is available in the ACS.

Despite these limitations, the findings consistently point to a labour market where the highest-paying arrangements — extreme hours, masculine sectors, uninterrupted careers — are systematically less remunerative for women, even after conditioning on everything observable. Whether this reflects discrimination, differential bargaining power, or the unequal distribution of household constraints remains an open question, and one that requires richer data and more credibly causal designs to answer.

Appendix

A Tables

Table 3: Double Post-Lasso Estimates (Extension 1)

	Variable	Estimate	Std.Error	t.value	p.value
1	femTRUE:married	-0.047***	0.010	-4.822	0.000
2	femTRUE:chld19	-0.045***	0.010	-4.516	0.000
3	femTRUE:regionEast South Central Div.	-0.005	0.019	-0.267	0.789
4	femTRUE:regionMiddle Atlantic Division	0.003	0.013	0.195	0.845
5	femTRUE:regionMountain Division	-0.010	0.017	-0.603	0.547
6	femTRUE:regionNew England Division	0.010	0.017	0.584	0.559
7	femTRUE:regionPacific Division	-0.013	0.014	-0.930	0.352
8	femTRUE:regionSouth Atlantic Division	0.015	0.013	1.176	0.240
9	femTRUE:regionWest North Central Div.	-0.014	0.017	-0.869	0.385
10	femTRUE:regionWest South Central Div.	-0.066***	0.015	-4.276	0.000
11	femTRUE:msa	0.035**	0.015	2.320	0.020
12	femTRUE:degBio/Life Sci	-0.057*	0.034	-1.689	0.091
13	femTRUE:degBus	-0.033	0.031	-1.083	0.279
14	femTRUE:degComm	0.019	0.033	0.565	0.572
15	femTRUE:degComp/Inform Sci	-0.084**	0.034	-2.473	0.013
16	femTRUE:degCrim Just/Fire Prot	-0.065*	0.035	-1.882	0.060
17	femTRUE:degEduc Admin/Teach	-0.031	0.033	-0.934	0.350
18	femTRUE:degEngin	-0.008	0.034	-0.239	0.811
19	femTRUE:degEnglish/Lit/Compos	0.013	0.034	0.386	0.699
20	femTRUE:degFine Arts	-0.040	0.034	-1.183	0.237
21	femTRUE:degHist	-0.086***	0.032	-2.674	0.007
22	femTRUE:degLib Arts/Hum	0.018	0.042	0.437	0.662
23	femTRUE:degMath/Stats	-0.035	0.038	-0.909	0.364
24	femTRUE:degMed/Hlth Sci Serv	-0.045	0.035	-1.267	0.205
25	femTRUE:degPhys Fit/Parks/Recr/Leis	-0.047	0.035	-1.326	0.185
26	femTRUE:degPhys Sci	-0.107***	0.032	-3.345	0.001
27	femTRUE:degPsych	-0.039	0.035	-1.114	0.265
28	femTRUE:degPubl Aff/Policy/Soc Wo	-0.092**	0.046	-1.986	0.047
29	femTRUE:degSoc Sci	-0.046	0.032	-1.422	0.155
30	femTRUE:occArts/Design/Entert/Sports/Media	-0.063**	0.029	-2.189	0.029
31	femTRUE:occBus Operat Spec	-0.084***	0.027	-3.074	0.002
32	femTRUE:occComm/Soc Serv	0.027	0.033	0.828	0.408
33	femTRUE:occComput/Math	-0.094***	0.028	-3.315	0.001
34	femTRUE:occEduc/Training/Libr	-0.023	0.028	-0.810	0.418
35	femTRUE:occFinanc Spec	-0.126***	0.028	-4.513	0.000
36	femTRUE:occHealthc Pract/Technic	-0.078***	0.029	-2.656	0.008
37	femTRUE:occLegal	-0.064**	0.032	-2.029	0.042
38	femTRUE:occLife/Physical/Soc Sci.	-0.020	0.032	-0.602	0.547
39	femTRUE:occManagmnt, Bus/Science/Art	-0.080***	0.025	-3.197	0.001
40	femTRUE:occOffice/Administr Supp	-0.087***	0.027	-3.239	0.001
41	femTRUE:occProd	-0.077*	0.041	-1.893	0.058
42	femTRUE:occProtect Serv	-0.078*	0.040	-1.956	0.050
43	femTRUE:occSales	-0.075***	0.027	-2.770	0.006
44	femTRUE:indAGRI	-0.009	0.047	-0.187	0.852
45	femTRUE:indBUISREPSERV	-0.034*	0.021	-1.654	0.098
46	femTRUE:indCONSTR	-0.031	0.033	-0.925	0.355
47	femTRUE:indENTER	-0.022	0.034	-0.644	0.520
48	femTRUE:indFINANCE	-0.065***	0.019	-3.389	0.001
49	femTRUE:indMANUF	0.011	0.020	0.549	0.583
50	femTRUE:indPERSON	-0.022	0.035	-0.633	0.527
51	femTRUE:indPROFE	-0.030*	0.017	-1.750	0.080
52	femTRUE:indRETAIL	0.023	0.021	1.096	0.273
53	femTRUE:indTRANS	-0.013	0.024	-0.569	0.569
54	femTRUE:indWHOLESALE	0.028	0.028	1.017	0.309
55	femTRUE:hw50to59	0.017*	0.010	1.770	0.077
56	femTRUE:hw60to69	-0.025	0.016	-1.590	0.112
57	femTRUE:hw70plus	-0.065**	0.027	-2.398	0.016
58	femTRUE:hwLessthan40	0.016	0.019	0.840	0.401
59	femTRUE:yos	0.039***	0.013	2.985	0.003
60	femTRUE:exp	0.025**	0.011	2.228	0.026
61	femTRUE:exp2	0.008	0.009	0.898	0.369
62	femTRUE:age	-0.037***	0.010	-3.638	0.000

Table 4: Causal Forest: Best Linear Projections (Extension 1)

	Variable	Estimate	Std.Error	t.value	p.value
1	(Intercept)	-0.033	0.080	-0.416	0.678
2	married	-0.042***	0.010	-4.229	0.000
3	chld19	-0.043***	0.010	-4.131	0.000
4	regionEast.North.Central.Div.	0.054***	0.016	3.320	0.001
5	regionEast.South.Central.Div.	0.065***	0.021	3.042	0.002
6	regionMiddle.Atlantic.Division	0.045***	0.017	2.682	0.007
7	regionMountain.Division	0.051***	0.019	2.606	0.009
8	regionNew.England.Division	0.073***	0.019	3.791	0.000
9	regionPacific.Division	0.033*	0.018	1.841	0.066
10	regionSouth.Atlantic.Division	0.072***	0.016	4.476	0.000
11	regionWest.North.Central.Div.	0.048**	0.019	2.515	0.012
12	msa	0.042***	0.016	2.689	0.007
13	degBio.Life.Sci	-0.028	0.041	-0.691	0.489
14	degBus	-0.025	0.038	-0.661	0.509
15	degComm	0.032	0.040	0.797	0.426
16	degComp.Inform.Sci	-0.067	0.046	-1.464	0.143
17	degCrim.Just.Fire.Prot	-0.028	0.045	-0.621	0.535
18	degEduc.Admin.Teach	-0.027	0.039	-0.694	0.488
19	degEngin	0.012	0.043	0.279	0.780
20	degEnglish.Lit.Compos	0.024	0.042	0.563	0.573
21	degFine.Arts	-0.023	0.041	-0.559	0.576
22	degHist	-0.035	0.044	-0.797	0.426
23	degLib.Arts.Hum	0.043	0.050	0.860	0.390
24	degMath.Stats	-0.054	0.051	-1.061	0.288
25	degMed.Hlth.Sci.Serv	-0.028	0.042	-0.663	0.508
26	degPhys.Fit.Parks.Recr.Leis	0.009	0.045	0.193	0.847
27	degPhys.Sci	-0.073*	0.044	-1.653	0.098
28	degPsych	-0.009	0.041	-0.216	0.829
29	degPubl.Aff.Policy.Soc.Wo	-0.087*	0.051	-1.718	0.086
30	degSoc.Sci	-0.032	0.040	-0.818	0.414
31	occArts.Design.Entert.Sports.Media	0.022	0.029	0.752	0.452
32	occBus.Operat.Spec	-0.008	0.026	-0.309	0.757
33	occComm.Soc.Serv	0.066**	0.029	2.290	0.022
34	occComput.Math	-0.019	0.026	-0.736	0.462
35	occEduc.Training.Libr	0.042*	0.024	1.737	0.082
36	occFinanc.Spec	-0.040	0.027	-1.511	0.131
37	occHealthc.Pract.Technic	0.034	0.028	1.209	0.227
38	occLegal	0.034	0.033	1.032	0.302
39	occLife.Physical.Soc.Sci.	0.042	0.032	1.342	0.179
40	occManagmnt..Bus.Science.Art	-0.010	0.022	-0.425	0.671
41	occOffice.Administr.Supp	-0.037	0.025	-1.446	0.148
42	occProd	0.080*	0.046	1.738	0.082
43	occProtect.Serv	0.023	0.037	0.640	0.522
44	occSales	-0.012	0.027	-0.467	0.640
45	indAGRI	-0.103**	0.051	-2.029	0.042
46	indBUISREPSERV	-0.015	0.021	-0.717	0.473
47	indCONSTR	0.020	0.031	0.636	0.525
48	indENTER	0.024	0.046	0.514	0.607
49	indFINANCE	-0.056***	0.020	-2.848	0.004
50	indMANUF	0.044**	0.020	2.250	0.024
51	indPERSON	-0.056	0.041	-1.383	0.167
52	indPROFE	-0.001	0.016	-0.061	0.951
53	indRETAIL	0.036	0.022	1.617	0.106
54	indTRANS	0.008	0.024	0.332	0.740
55	indWHOLESALE	0.053*	0.030	1.770	0.077
56	hw50to59	0.015	0.010	1.415	0.157
57	hw60to69	-0.027	0.021	-1.315	0.189
58	hw70plus	-0.209***	0.039	-5.352	0.000
59	hwLessthan40	-0.018	0.020	-0.902	0.367
60	yos	-0.004	0.004	-1.124	0.261
61	exp	0.001	0.004	0.257	0.797
62	exp2	-0.006	0.010	-0.567	0.571

Table 5: Causal Forest: Feature Importance (Extension 1)

	Variable	Importance
1	married	0.15
2	nevermarried	0.15
3	chld19	0.20
4	regionEast North Central Div.	0.00
5	regionEast South Central Div.	0.00
6	regionMiddle Atlantic Division	0.00
7	regionMountain Division	0.00
8	regionNew England Division	0.00
9	regionPacific Division	0.00
10	regionSouth Atlantic Division	0.01
11	regionWest North Central Div.	0.00
12	regionWest South Central Div.	0.01
13	msa	0.00
14	degBio/Life Sci	0.00
15	degBus	0.01
16	degComm	0.02
17	degComp/Inform Sci	0.00
18	degCrim Just/Fire Prot	0.00
19	degEduc Admin/Teach	0.00
20	degEngin	0.00
21	degEnglish/Lit/Compos	0.00
22	degFine Arts	0.00
23	degHist	0.00
24	degLib Arts/Hum	0.00
25	degMath/Stats	0.00
26	degMed/Hlth Sci Serv	0.00
27	degPhys Fit/Parks/Recr/Leis	0.00
28	degPhys Sci	0.00
29	degPsych	0.00
30	degPubl Aff/Policy/Soc Wo	0.00
31	degSoc Sci	0.01
32	occArts/Design/Entert/Sports/Media	0.01
33	occBus Operat Spec	0.01
34	occComm/Soc Serv	0.00
35	occComput/Math	0.00
36	occEduc/Training/Libr	0.00
37	occFinanc Spec	0.02
38	occHealthc Pract/Technic	0.00
39	occLegal	0.00
40	occLife/Physical/Soc Sci.	0.00
41	occManagmnt, Bus/Science/Art	0.01
42	occOffice/Administr Supp	0.01
43	occProd	0.00
44	occProtect Serv	0.00
45	occSales	0.01
46	indAGRI	0.00
47	indBUISREPSERV	0.00
48	indCONSTR	0.00
49	indENTER	0.00
50	indFINANCE	0.15
51	indMANUF	0.01
52	indPERSON	0.00
53	indPROFE	0.00
54	indRETAIL	0.01
55	indTRANS	0.00
56	indWHOLESALE	0.00
57	hw50to59	0.01
58	hw60to69	0.01
59	hw70plus	0.00
60	hwLessthan40	0.00
61	yos	0.01
62	exp	0.04
63	exp2	0.05
64	age	0.05

B References

References

Susan Athey and Stefan Wager. Estimating treatment effects with causal forests: An application. *Observational Studies*, 5(2):37–51, 2019. URL https://muse.jhu.edu/article/793356.

David Averkamp, Christian Bredemeier, and Falko Juessen. Decomposing gender wage gaps: A family economics perspective. *The Scandinavian Journal of Economics*, 126(1):3–37, 2024. doi: 10.1111/sjoe.12539.

Philipp Bach, Victor Chernozhukov, and Martin Spindler. Closing the us gender wage gap requires understanding its heterogeneity. *arXiv preprint arXiv:1812.04345*, 2018. URL https://arxiv.org/abs/1812.04345.

Philipp Bach, Malte S. Kurz, Victor Chernozhukov, Martin Spindler, and Sven Klaassen. DoubleML: An object-oriented implementation of double machine learning in R. *Journal of Statistical Software*, 108(3):1–56, 2024. doi: 10.18637/jss.v108.i03.

Alexandre Belloni, Victor Chernozhukov, and Christian Hansen. Inference on treatment effects after selection among high-dimensional controls. *Review of Economic Studies*, 81(2):608–650, 2014a. doi: 10.1093/restud/rdt044.

Alexandre Belloni, Victor Chernozhukov, and Kengo Kato. Uniform post-selection inference for least absolute deviation regression and other z-estimation problems. *Biometrika*, 102(1):77–94, 2014b. doi: 10.1093/biomet/asu071.

Marianne Bertrand, Claudia Goldin, and Lawrence F. Katz. Dynamics of the gender gap for young professionals in the financial and corporate sectors. *American Economic Journal: Applied Economics*, 2(3):228–255, 2010. doi: 10.1257/app.2.3.228.

Francine D Blau and Lawrence M Kahn. The gender wage gap: Extent, trends, and explanations. *Journal of Economic Literature*, 55(3):789–865, 2017.

Michelle J. Budig and Paula England. The wage penalty for motherhood. *American Sociological Review*, 66(2): 204–225, 2001.

Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. *Annals of Statistics*, 41(6):2786–2819, 2013a. doi: 10.1214/13-AOS1161.

Victor Chernozhukov, Denis Chetverikov, and Kengo Kato. Gaussian approximation of suprema of empirical processes. *Annals of Statistics*, 42(4):1564–1597, 2014. doi: 10.1214/14-AOS1234.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097.

Markus Gangl and Andrea Ziefle. Motherhood, labor force behavior, and women's careers: An empirical assessment of the wage penalty for motherhood in britain, germany, and the united states. *Demography*, 46(2):341–369, 2009.

Claudia Goldin. A grand gender convergence: Its last chapter. *American Economic Review*, 104(4):1091–1119, 2014. doi: 10.1257/aer.104.4.1091.

Joseph P. Romano and Michael Wolf. Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4): 1237–1282, 2005a. doi: 10.1111/j.1468-0262.2005.00615.x.

Joseph P. Romano and Michael Wolf. Exact and approximate stepdown methods for multiple hypothesis testing. *Journal of the American Statistical Association*, 100(469):94–108, 2005b. doi: 10.1198/016214504000000539.

Joseph P. Romano and Michael Wolf. Efficient computation of adjusted p-values for resampling-based stepdown multiple testing. *Statistics & Probability Letters*, 113:38–40, 2016. doi: 10.1016/j.spl.2016.02.012.

U.S. Bureau of Labor Statistics. The employment situation — january 2024. Technical report, U.S. Department of Labor, February 2024. URL https://www.bls.gov/news.release/empsit.nr0.htm. Accessed: 2026-03-27.

Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018. doi: 10.1080/01621459.2017.1319839.

Jane Waldfogel. The effect of children on women's wages. *American Sociological Review*, 62(2):209–217, 1997.